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COS - AT.1

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1) Parallelism Degree Calculation - width of a DAG

Idea:-

The width is the maximum number of tasks that can be executed simultaneously, meaning no task in that group depends on another task in the same group.

Pseudocode:-

ALGORITHM FindWidth(DAG)

Input:- DAG $G = (V, E)$

Output:- Width of the DAG

1. Find all possible sets of tasks where.

2. $\text{Maxwidth} \leftarrow 0$.3. For each possible set S of tasks;

IF no dependency exists between any two tasks in S :

Size $\leftarrow$ number of tasks in S

IF Size $>$ maxwidth:

maxwidth $\leftarrow$ size

Return maxwidth

2) Topological Sort on a Forest.

Here we have several independent DAGs/ components. We need to provide a order while trying to execution different components

Pseudocode:-

ALGORITHM

Alternate Topological Sort (Forest)

Input: A Forest containing several DAGs

Output:- Valid execution order.

1. For each component C :

Find the indegree of every task.

Add all tasks with indegree of every

2. Create an empty list result.

3. While there are unprocessed tasks:

For each component C in ground-robin order:

IF ready[C] is not empty:

task ← remove a task
from ready[C].

Add task to result

For each task T dependent on task:

Decrease

indegree[T]

IF indegree[T] = 0:

Add T to

ready[C]

4. Return result

3/18/20